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Climate Change Adaptation Knowledge Among Rice Farmers in Lake Toba Highland, Indonesia

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Abstract

Climate change has increasingly disrupted traditional farming systems, particularly in highland areas where environmental changes are more pronounced. This study explores how rice farmers in the Lake Toba highlands, Indonesia—both irrigated and non-irrigated—have gradually shifted away from traditional knowledge (TK) in response to climate challenges and what new adaptation strategies have emerged to sustain rice production. This study employed a descriptive qualitative approach with a broad and holistic perspective. Data were collected from 130 purposively selected rice farmers in two sub-districts: Harian (irrigated) and Pangururan (non-irrigated). Data were gathered through in-depth interviews guided by semi-structured statements, focusing on farmers' lived experiences and adaptation strategies across the rice farming cycle—from planting to harvesting. The findings revealed that while the two groups differ in water access and environmental conditions, they show similar trends in shifting away from traditional indicators. Farmers increasingly adopted new adaptation strategies such as joining farmer groups, using water pumps in non-irrigated areas, switching to more climate-resilient crop varieties, and adjusting planting calendars based on personal observation rather than inherited natural signs. This shift from traditional to practical, experience-based strategies reflects farmers' responses to the fading reliability of traditional knowledge under changing climatic conditions. Despite the loss of symbolic TK practices, farmers continue to demonstrate resilience through peer collaboration and contextual decision-making. This study highlights the need to strengthen farmer-led adaptation while preserving valuable elements of TK. Future research should expand across the Lake Toba highlands and incorporate quantitative methods to capture broader patterns of local adaptation.

Keywords: climate change; traditional farmers' knowledge; Lake Toba; irrigation farmers; non-irrigation farmers; highland paddy farmers

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1. Introduction

In Indonesia, climate change poses a significant risk of increasing the poverty rate. According to data from March 2022, 26.16 million people, accounting for 9.54% of the total population, are living below the poverty line. About 67% of the population is in the vulnerable position of poverty. This category of people are neither classified as poor

nor middle class but risk falling into deprivation when affected by climate change [1–4]. The agricultural sector, specifically farmers and the public as consumers, are at the risk of declining commodity production. In 2023, the total financial loss due to climate change reached IDR 115.53 trillion, with losses in the agricultural sector reaching IDR 19.94 trillion. These losses include 27 million household farmers, including 17 million rice farmers with an average land ownership of 0.6 ha [5]. In general, household farms in Indonesia are undereducated, adopting and using inefficient technologies [6].

Rice is an important food crop widely cultivated in Indonesia but highly affected by climate change [7,8]. Consequently, rice farmers carry out adaptation strategies as a location-specific global phenomenon [9]. In the tropics, the main factors that influence farm households decisions to adapt to climate change include [10], a lack of access to agricultural extension services [11], access to climate information [12–14], changes in rainfall patterns [14,15], and yield loss [16,17]. Climate change impacts tropical plants due to increased temperature stress [18]. Many adaptation strategies at the farm level entail adjustment to planting and harvesting dates [19], crop rotation, quality seed selection [20,21], the selection of crops and crop varieties for cultivation, water consumption for irrigation [22], fertilizer use, and tillage practices [23]. The natural adaptation results from producers' goal to maximize returns to land resources. Each adaptation reduces potential yield losses due to climate change or increases yields where climate change can be used [24–26].

Adaptation strategies related to climate change have been widely discussed, with a focus on economic incentives [26,27], overcoming negative impacts on various crop production [28,29]. Examining adaptation strategies in agriculture to climate change is important to increase adaptive capacity by empowering communities and improving ecosystem functionality [30]. Various adaptation strategies to sustain crop yields have been developed in different countries, such as fertilizer application, the planting of adaptive crop varieties, and using improved irrigation methods or water collection technologies in Uganda. Agronomic adaptation measures include early planting, mulching, terrace creation, contour plowing, land emptying, shifting cultivation, the use of compost fertilizer, intercropping, and crop rotation. Off-farm adaptive measures include migration, loans, and taking off-farm jobs [31] or diversifying crops from rice to cassava, oil palm, sugarcane, mango, watermelon, and vegetables, changing planting calendars and crop varieties, increasing the use of farm machinery, or shifting planting locations. Farmers have formed cooperatives to implement these adaptation measures due to the cost for individuals [32].

Meanwhile, the application of adaptation strategies involving farmers' social participation and addressing the impact of climate change on the culture and traditional knowledge (TK) of farming communities remains limited, including in the context of different irrigation systems. This study examined differences between rice farmers in hilly areas who use irrigation systems as a source of water for farming (irrigation farmers) and those in the lowland areas near Lake Toba who do not use formal irrigation systems (non-irrigation farmers). Despite being located within the same location in Lake Toba, these two groups differ in terms of agricultural practices and access to water sources, which affects their awareness, behaviors, and adaptation strategies to climate change—from planting to harvesting. To strengthen the analytical foundation, this study specifically selected these two farmer groups based on ecological and infrastructural distinctions. Harian Sub-District represents a hilly terrain where farmers utilize a gravity-based irrigation system sourced from the Efrata Waterfall. In contrast, Pangururan Sub-District comprises flatter lowland areas where farmers rely on rainfall or pump water from Lake Toba without permanent irrigation infrastructure. These physical differences shape the farmers' exposure to climate-related risks and their adaptive capacity.

In addition, the study highlights the role of traditional knowledge (TK) systems in climate adaptation. Local farmers have historically relied on TK—such as interpreting seasonal changes through wind direction, animal behavior, or other natural indicators—to guide agricultural activities. However, with increasingly unpredictable climate conditions, these traditional cues have become less reliable. As a result, many farmers have gradually shifted toward experience-based adaptation strategies, such as crop diversification, changing planting schedules and relying more heavily on farmer groups rather than inherited knowledge systems. These contextual differences and the evolving nature of TK use form the basis of this study's objective: to explore how two distinct farming communities have adapted to climate change and how their traditional knowledge systems have transformed in response to environmental uncertainty.

2. Methods

2.1. Location

This study was conducted in Samosir Regency, North Sumatra Province, Indonesia, focusing on two sub-districts: Harian and Pangururan (Figure 1). Geographically, the research area is located between 2°01′38″–2°49′48″ North Latitude and 98°24′00″–99°01′48″ East Longitude, with an elevation ranging from approximately 904 to 2157 m above sea level, covering an area of about ±2069.05 km^2. Harian Sub-District is situated at an altitude of approximately 2157 m above sea level and is characterized by hilly terrain that directly faces Lake Toba. The region is surrounded by highlands, forming a closed valley with complex wind circulation patterns. Winds often swirl from the hills toward the lake and vice versa. This geographical configuration contributes to high rainfall intensity and frequent strong winds, which significantly affect agricultural activities. One of the key natural resources in this area is the Efrata Waterfall, which serves as the primary source of irrigation for rice fields.

Figure 1. Research location map (Pangururan District And Harian District), North Sumatera, Indonesia.

In contrast, Pangururan Sub-District lies at an altitude of around 904 m above sea level. The agricultural landscape in this area varies, with some villages located near the lake, while others are situated farther inland. Compared to Harian, Pangururan experiences relatively low rainfall, and farming activities rely heavily on the rainy season. During extended dry periods, farmers turn to water pumping systems that draw from Lake Toba as an alternative irrigation method. Both areas are known for their communities that maintain traditional ways of life, particularly in agriculture [33], tourism [34,35], forest, education, health practices [36,37], and daily social activities. Local cultural values and traditional knowledge continue to play a vital role in how these communities manage natural resources and respond to environmental changes, including the increasingly unpredictable impacts of climate change.

2.2. Informants

The number of informants in this study was set at 130 people, with a balanced distribution between Harian Sub-District and Pangururan Sub-District, with 65 farmers from each sub-district. This number was determined based on methodological considerations and field context, using a purposive sampling approach, which involves the deliberate selection of informants (non-random) based on specific criteria relevant to the research objectives. The use of purposive sampling in qualitative research aims to ensure that the selected informants have knowledge, experience, and direct involvement with the phenomenon being studied—in this case, farmers' adaptation to climate change based on their farming systems (irrigated and non-irrigated).

The criteria used for selecting informants include the following:

1. Farmers who have agricultural land adjacent to Lake Toba (both in highland and lowland areas);
2. Those who have at least 20–30 years of farming experience;
3. Those who are members of an active farmer group.

The selection of 65 informants per sub-district was made to allow for a sufficient variation in experiences and perspectives within a socially homogeneous area while maintaining the depth of data exploration qualitatively. Furthermore, this sample size was chosen to ensure that the data collected would provide a holistic understanding of the research topic. While the number of respondents in each sub-district was relatively large, it was important to capture diverse perspectives on the adaptation strategies employed by farmers in both irrigated and non-irrigated farming systems. Additionally, this sample size ensures that this study reaches data saturation, where no new information is emerging from the interviews, thereby validating the comprehensiveness and richness of the data.

In qualitative research, statistical representativeness is not the primary goal but rather the depth, relevance, and diversity of the information obtained [38,39]. Additionally, the even distribution between the two sub-districts allows for a contextual comparison between two groups of farmers with different geographical characteristics and farming systems. Therefore, the number and strategy for selecting informants are considered adequate to support contextual validity and the narrative strength of the data in the qualitative approach.

2.3. Data Collection

Data were collected through in-depth interviews using a set of semi-open-ended questions. A total of 45 questions were asked, covering various aspects such as land ownership, knowledge of climate change and agriculture, adaptation strategies, available resources, and local leadership. The interviews were conducted by trained enumerators who had undergone a preparatory training phase prior to data collection.

As part of the training, enumerators were authorized to adapt the language and wording of the provided questions to better suit the local context, given that the majority of respondents were members of the Batak Toba ethnic group. To ensure clarity and cultural relevance, enumerators were encouraged to simplify the questions—without altering their intended meaning—so that informants could more easily understand them. Additionally, enumerators were instructed to use probing techniques when respondents' answers were unclear or required further elaboration. This approach allowed for more meaningful and contextually grounded data collection.

All interviews were conducted verbally in accordance with ethical research standards, taking place at the respondents' homes or agricultural fields to allow for flexible and natural interactions. Prior to each interview, verbal consent was obtained, ensuring that participation was entirely voluntary. Informants had the right to decline participation at any point, and their decisions were fully respected by the research team. Verbal consent was documented by the enumerators as part of the ethical protocol.

2.4. Data Analysis

The data in this study were analyzed using a thematic analysis approach, which is commonly employed in qualitative research to identify patterns of meaning (themes) across a dataset. The process involved organizing, interpreting, and synthesizing responses from both the semi-open-ended questionnaires and in-depth interviews with a total of 130 informants—65 farmers from Harian Sub-District and 65 from Pangururan Sub-District. While several frequencies and simple percentages were used to describe general trends in responses (e.g., how many farmers reported crop failure or used water pumps), the emphasis of the analysis remained qualitative. The use of these numerical illustrations served only to support thematic interpretations, not to perform statistical inference. This approach aligns with the qualitative tradition, where the goal is to understand meaning, context, and lived experiences, rather than to generalize findings statistically.

Additionally, responses to the closed-ended questions were processed using SPSS 26 software to calculate the frequency distribution of each item, allowing for a clearer presentation of how responses were distributed among informants. In some cases, responses to open-ended questions that showed relatively homogeneous patterns were also entered into SPSS to generate supportive frequency data. However, the use of SPSS in this study was limited to descriptive purposes only, without proceeding to hypothesis testing. This integration of SPSS-assisted frequency analysis aimed to enrich the narrative and descriptive aspects of the qualitative findings, thereby producing more varied and holistic insights.

3. Results and Discussion

3.1. Climate Change Awareness

Irrigated and non-irrigated rice farmers' awareness of climate change was measured based on knowledge of (a) the occurrence of extreme situations over the last 10 years, (b) rainfall, (c) water discharge in rivers, and (d) air temperature (Table 1). The extreme situation in question is a prolonged dry season characterized by drought and frequent strong winds. This situation was perceived by 43 irrigated rice farmers (66.1%) and 64 non-irrigated rice farmers (98.5%). A decreasing trend in rainfall was observed by 47.8% of irrigated and 93.9% of non-irrigated farmers. Additionally, 33.8% of irrigated and 84.6% of non-irrigated farmers reported a continuous decline in river water discharge, affecting the irrigation of rice fields. An increase in air temperature in the last 10 years was observed by 37.0% of irrigated and 52.3% of non-irrigated farmers.

Table 1. Local community awareness of climate change.

Statements		Paddy Planting	
		Irrigation	Non–Irrigation
Extreme situations that have been experienced	Drought	43 (66.1%)	64 (98.5%)
	Heavy rains	3 (4.7%)	-
	High winds	15 (23.0%)	-
	Not experienced	4 (6.2%)	1 (1.5%)
Rainfall conditions in this area over the last 10 years	Unchanged	19 (29.2%)	2 (3.1%)
	Uncertain	9 (13.8%)	1 (1.5%)
	Decrease	31 (47.8%)	61 (93.9%)
	Increase	6 (9.2%)	1 (1.5%)
The state of river water discharge used for irrigation over the past 10 years	Unchanged	26 (40.0%)	3 (4.7%)
	Uncertain	13 (20.0%)	6 (9.2%)
	Decrease	22 (33.8%)	55 (84.6%)
	Increase	4 (6.2%)	1 (1.5%)
Air temperature conditions in this area over the last 10 years	Unchanged	16 (24.6%)	28 (43.0%)
	Uncertain	18 (27.7%)	-
	Decrease	7 (10.7%)	3 (4.7%)
	Increase	24 (37.0%)	34 (52.3%)

Source: authors' own work, 2024.

In general, rice farmers' views on climate change were only related to (a) the occurrence of extreme rains and (b) the incident of extreme winds. Extreme rain is characterized by heavy rain and subsequent intense heat. Furthermore, when frequent 'sunshower' (locally known as "*singgar*") occurs, rain and heat co-occur. This situation has a terrible impact on rice crop. Due to these two weather conditions, rice farmers often experience a situation where rice stalks gradually turn yellow. However, the paddy seeds may remain empty or contain smelly caterpillars. Extreme wind is characterized by the sudden arrival of strong winds at night that break the stalks of rice or corn stalks and only occur during the prolonged dry season. Both irrigated and non-irrigated farmers have a relatively high awareness of the climate change that has occurred over the past decade. The majority of respondents from both groups stated that air temperature has increased, rainfall has tended to decrease, and river water discharge has also shown a decline. This condition has been particularly felt more strongly by non-irrigated farmers, who are generally more affected by climate change. The most significant extreme impact experienced by the community is drought, while other impacts such as heavy rainfall and strong winds were only felt by a small portion of the respondents. This awareness shows that climate change is no longer just a global issue but has become a real experience affecting the daily lives of the local community, particularly in the agricultural sector.

Climate change is interpreted when extreme events apply to agricultural produce. In other parts of the world, as observed by Petersen-Rockney Margiana [40], there is still a gap between the scientific understanding of climate change and farmers' adoption of adaptation and mitigation practices [41]. Many US farmers do not believe in anthropogenic climate change [42], do not consider climate change a local risk to farming operations [43], and have not implemented management practices that reduce emissions or foster resilience [44]. In addition, farmers often distrust public agricultural advisors such as Cooperative Extension staff who discuss climate change [45]. These findings align with the observations in the Lake Toba highlands, where farmers' awareness and understanding of climate change are also shaped by direct experiences rather than scientific explanations. Like U.S. farmers, the farmers in this study tend to base their adaptive decisions on personal observations of weather anomalies rather than broader scientific discourse. However, unlike their U.S.

counterparts, most farmers in this study do acknowledge that climate patterns are changing, even if they lack access to formal education or technical climate data. This reinforces the idea that both local knowledge and perception play a crucial role in shaping agricultural responses to climate challenges across different contexts.

3.2. Behavior of Irrigated and Non-Irrigated Rice Farmers in Facing Climate Change

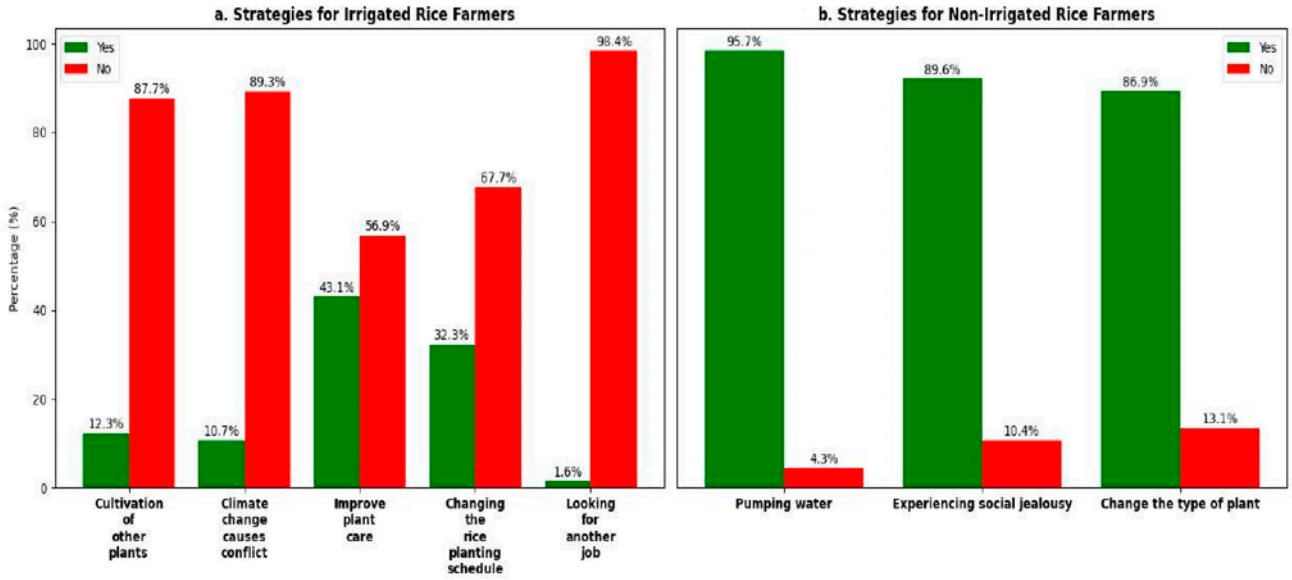
The behavior of rice farmers towards climate change is reflected in (a) adaptation strategies of irrigated farmers living in mountainous areas, (b) adaptation strategies of non-irrigated farmers in the lowlands, and (c) irrigated and non-irrigated rice farmers who carry out adaptation strategies based on their experience of dealing with climate change. Behavioral differences are based on variations in the location of rice planting land and irrigation facilities on farms. However, the experience in dealing with climate change was the same. The differences and similarities of adaptation strategies to climate change in these two types of rice farmers are described as follows.

3.2.1. Adaptation Strategies of Irrigated Rice Farmers

Irrigated rice farmers fully depend on farming for livelihoods and economic activities, making it difficult to change the type of crop. For the Toba Batak people, rice has a distinct cultural value, not only as the main food but also a source of family honor [46,47].

Scheme 1 shows that only 12.3% of irrigated farmers were willing to replace rice with more economically valuable crops when the dry season was prolonged. Farmers who adopted another crop type believed that replacing rice with other crops would improve and increase soil fertility. This will also impact the quality of rice planted in the subsequent season [48,49]. Another motivation is that diversifying crops reduces the impact of economic losses on farmers. "In several cases, farmers with larger agricultural plots engage in crop diversification by cultivating rice alongside other food crops such as corn, which is known for its resilience to weather fluctuations. Corn is particularly favored due to its shorter growing cycle, enabling quicker returns and providing timely financial support for household needs". This strategy helps reduce the enormous losses incurred when extreme weather changes occur. When rice crops experience failure, farmers will not encounter a significant loss due to the presence of corn or other crops as alternatives [50,51]. "Farmers' choice of alternative crops was primarily informed by their accumulated experiences rather than by formal knowledge of soil characteristics or ecological compatibility. These decisions reflect a practical orientation toward crop varieties that have proven to withstand the impacts of climate change".

Conflicts over water sources occurred during the prolonged dry season, as mentioned by 10.7% of irrigated rice farmers. Irrigation is used as the artificial channeling or delivery of water to ensure its availability and supply throughout the year. It is an important technique farmers use to adapt to climate change [52]. The primary water source for irrigated rice fields in the study area is the Efrata Waterfall, which stands approximately 26 m high and 12 m wide. While this waterfall has never experienced complete drought, its water discharge significantly decreases during prolonged dry seasons. This decline in water volume leads to competition among farmers to access limited water for their fields. Farmers whose rice fields are situated at higher elevations tend to have better access to water, while those located at lower altitudes often suffer from water shortages. As a result, the scarcity of water not only threatens crop health—leading to increased pest attacks—but also sparks conflicts over water usage.



Scheme 1. Rice farmers' behavior in facing climate change. Source: authors' work, 2024.

These conflicts typically involve farmers attempting to redirect the flow of water from the waterfall to their own land. In many cases, farmers guard the water source at night and manually divert water to their rice fields. If left unguarded, other farmers may reroute the flow to their own fields. This recurring struggle for water access during drought periods is the root cause of many conflicts among irrigated rice farmers in the region. Despite mitigation efforts such as fertilizer application, pest control, and intensive field management—reported by 43.1% of irrigated farmers—yields often remain suboptimal due to the environmental and social disruptions caused by these water shortages [53].

The choice to change the rice planting schedule was only made by 32.3% of irrigated rice farmers in the event of extreme climate change. The schedule change was only implemented by rice farmers whose paddy fields were at the foot of the mountains due to the fear of insufficient water.

Accommodating climate change by changing planting and harvesting dates is not fully applicable [54] for irrigated farmers in this region, and rice varieties also need to be considered [55]. Switching to other occupations for irrigated rice farmers is possible, but leaving this occupation completely is rare. Only 4.6% intend to switch jobs but will return to rice farming when extreme droughts subside.

3.2.2. Adaptation Strategies of Non-Irrigated Rice Farmers

The results on adaptation strategies of non-irrigated rice farmers are presented in Scheme 1b. Based on the results, about 95.7% of non-irrigated rice farmers conducted pumping to irrigate farmlands. Water pumps are an alternative used for rice fields located close to Lake Toba during the dry season. Lake Toba machines and pipes water flows directly into farmers' agricultural land. However, to use pumps, non-irrigated rice farmers must pay a rent of 30,000 IDR per hectare. The length of time to use the pump depends on the size of the field, usually taking more than 12 h. The high cost and disproportionate yields have led to switching jobs as farm laborers elsewhere or leaving land fallow for a season until the following rainy season. Climate change is the biggest challenge farmers face with non-irrigated farming systems, known as *sabah langit* by the community.

When the farmland is located in a hilly area far from Lake Toba, non-irrigated farmers cannot use a pump, leaving rainfall as the only option. In Pangururan Sub-District, the location of farmers' land varies—some are situated near Lake Toba, while others are located in highland or hilly areas far from any accessible water sources. This topographical

variation makes it nearly impossible for farmers whose fields are in the hills to use water pumps for irrigation (Figure 2). As a result, these farmers are entirely dependent on daily rainfall or are forced to adapt by switching to alternative crops that are more resistant to drought and changing weather conditions. According to the Indonesian Ministry of Agriculture's 2024 records, rain-fed rice fields amount to 2.7 million hectares accounting for 36% of the 7.5 million hectares of rice fields. This implies that conjunctive water management, integrating different water sources, including harvested rainwater, treated wastewater, desalinized water, and groundwater, is vital in sustainable water resource management [56]. In the context of agriculture, inequality in access to water resources can lead to social jealousy, specifically among non-irrigated farmers. Social jealousy arises among non-irrigated farmers whose agricultural lands are located near Lake Toba and those farming in the hilly areas further away from the lake. Farmers with land close to the lake still have the opportunity to utilize surface water sources, such as by using pumps to irrigate their fields (Figure 2), especially during the dry season. In contrast, farmers in upland areas do not have access to such water sources due to geographical constraints that make the use of water pumps impossible. This situation creates a disparity in the capacity to adapt to climate change, particularly during droughts. Farmers near the lake are still able to grow rice year-round, while those in the hills are often forced to delay planting, switch to alternative crops, or leave their fields fallow. As a result, a sense of injustice and social jealousy emerges, as both groups are categorized as non-irrigated farmers, yet they face very different realities in terms of access to water resources. This condition underscores the importance of policy interventions that are more geographically sensitive and responsive to differences in natural resource accessibility, ensuring that climate change adaptation efforts are carried out more fairly and equitably across agricultural areas.

Figure 2. Pumping equipment in rice fields.

This inequality causes around 89.6% of non-irrigated farmers to experience social jealousy, which is rooted in differences in water access due to the geographical location of farms. When not managed properly, this situation has the potential to trigger conflict between farmers and reduce overall agricultural productivity. Therefore, more equitable and inclusive management strategies, such as the development of rainwater harvesting technologies and improved distribution infrastructure, are crucial in creating a balance in the utilization of water resources in the agricultural sector.

With the occurrence of extreme climate change, 86.9% of non-irrigated rice farmers made changes to short-term crops, which have a planting-to-harvest cycle of 3–4 months, such as corn (*Zea mays*), long beans (*Phaseolus vulgaris*), and ginger (*Zingiber officinale Rosc.*). Subsequently, farmers returned to planting rice when the rainy season returned. In this

way, some farmers experienced an increase in agricultural yields, as observed from last year's harvest, which reached 150 cans, and the last harvest of 200 cans. The harvests were sold and partly used for family consumption. The above types of plants can be grown on rain-fed land using multiple cropping or agroforestry systems between perennial crops [57], improving the economy of farming families [58].

3.2.3. Adaptation Strategies of Irrigated and Non-Irrigated Rice Farmers Based on Shared Experiences in Facing Climate Change

Table 2 shows the similarities between irrigated and non-irrigated rice farmers' experiences with climate change. The shared experiences were further analyzed based on each rice farmer's unique habits and TK including responses to crop failures caused by climate change. Both types of farmers show a similar experience as 72 (55.4%) have experienced crop failure, while 58 (44.6%) have not. However, there were differences in the experience. For irrigated rice farmers, crop failure was associated with decreased productivity. Despite a decrease in harvests compared to the previous year, there were still rice yields. For non-irrigated farmers, crop failure means not obtaining any harvest at all. Crop failure in irrigated rice farmers was caused by (1) strong winds that cause rice plants to bend or lodge, (2) continuous rain, leading to excessive water volume, which inhibits grain retention and promotes pest infestation and mold growth, and (3) prolonged drought, causing rice plants to become burned and wither. Meanwhile, for non-irrigated farmers, crop failure was often caused by a prolonged dry season. This often forces farmers to change their rice planting schedule and crops and even not plant for the year when there is no sign of rain. In irrigated farming systems, crop failure typically results in decreased productivity, meaning that while yields are reduced, some rice plants may still survive despite unfavorable conditions. This is due to the availability of water from irrigation, which mitigates the impact of irregular rainfall patterns. In contrast, non-irrigated farmers experience a more severe form of crop failure, where entire rice crops may be lost. Extended dry spells lead to crops being scorched and unable to grow or survive, as there is no alternative water source available. Thus, for non-irrigated farmers, the consequences of crop failure are more drastic, often resulting in the complete loss of the harvest, while irrigated farmers may only face a reduction in yield due to growth disruptions caused by excessive rainfall.

The results from the two areas showed that 80 respondents (61.5%) were not aware of wind-resistant rice seedling varieties, and only 50 other respondents (38.5%) were aware. Some rice varieties that local farmers use include (a) '*siserang*', with a short stem and capable of withstanding strong winds, (b) '*sibandung*', which are more resistant to strong winds due to the presence of elastic stems, (c) '*Sirambu Manis*', which only grows to thigh or adult height, and (d) '*Sisiopol*', which has the ability to resist drought. Some farmers have used local rice seeds for generations, but some change the seeds at every planting period.

The majority of rice farmers, estimated at 91.5%, have joined farmers' groups, primarily to access low-priced fertilizer, good rice seeds, and the organization program. Farmers' groups in the community routinely distribute fertilizer to members because it is difficult to receive from the government regularly. When the planting season arrives, but farmers fail to acquire fertilizer, the schedule that should enter the fertilization stage will be delayed. This subsequently impacts the timing of rice maturation and harvesting. About 71.4% of farmers stated that their farmer group has never shared information on climate change with its members. This farmers' group is a gathering place and an information center for members, particularly regarding the time to plant rice. Farmers also never received information or socialization about climate change from the local government. About 87.7% planted rice without relying on climate change information. Although 91.9% were aware of changes in agricultural productivity due to climate change, farmers argued that the causes include rising temperatures (14.3%), a lack of irrigation facilities (5.7%), poor-quality seeds

(1.9%), the cost of fertilizer (35.2%), a lack of equipment (2.9%), pest problems (14.3%), a lack of labor (6.7%), changeable weather (12.4%), and long drought (6.7%). These data show that climate change is not the absolute cause of the decline in rice yields, but other factors are involved.

Table 2. Behavior of irrigated rice farmers and non-irrigated rice farmers in facing climate change.

Adaptation Strategies of Irrigated and Non-Irrigated Rice Farmers	The Percentage
Have experienced crop failure due to climate change	72 (55.4%)
Farmers are not aware of low-emission rice varieties	80 (61.5%)
Joining farmers' groups	119 (91.5%)
Farmers' groups do not share information on climate change	93 (71.4%)
Farming systems no longer depend on climate change information	114 (87.7%)
Making changes to the harvest period due to erratic and sudden changes in the weather	70 (53.8%)
Changes in agricultural productivity	119 (91.1%)
Loss of farmers' traditional knowledge related to rainy and dry season schedules	126 (96.7%)

Source: authors' work, 2024.

3.3. Erosion of TK of Rice Farmers

Without information on climate change from the government or farmers' groups, rice farmers depend on TK to guide agricultural activities. However, this reliance is increasingly eroded due to climate change. The rainy season is a long-awaited and the most suitable time for planting rice, as seedlings will not lack water. Rice farmers are aware that August to December every year is the approximate time for the rainy season and the right time to start planting, while January to July is the dry season. This guideline has become local knowledge for the community.

Traditionally, several natural indicators have been used to predict the onset of the rainy season. Among other indicators is the direction of the wind from Lake Toba blowing north towards the hilly area, often accompanied by a sound. Another sign is the appearance of many fish in the waters. However, the community has felt the occurrence of climate change, and signs of TK about the schedule of the rainy and dry seasons have started to change. About 96.7% of farmers stated that the weather was no longer as predictable. For instance, planting rice traditionally starts in September, but the rains do not arrive until October, leading to the postponement of the planting process. Climate change has led to planting once a year, compared to the previous rule of twice a year.

Rice plant maintenance activities (Figure 3) start from the fertilization stage (*manganpui*) to protection from pests and animals (*mamuro*). In this process, rice farmers do not depend on climate change but use TK. For example, farmers dry rice fields when fertilizing to ensure the fertilizer is not wasted following the flow of water down and back into the fields. Other activities are carried out to protect rice from animal pests and birds, as well as weeding, to make rice ready for harvest with the characteristics of seeds, leaves, and rice stems already appearing yellow.

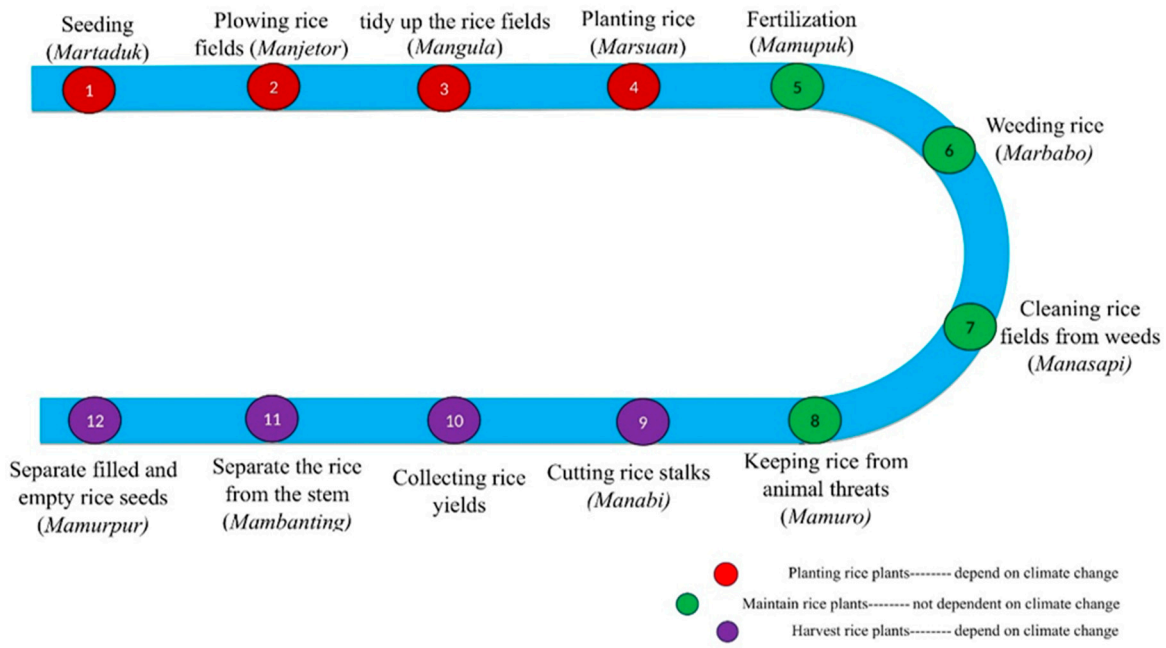


Figure 3. Stages of rice cultivation and dependency on climate change.

The most critical adaptation strategy carried out by farmers was to guard rice (*mamuro*) before the harvest period arrived. This process entails several protective measures to mitigate the risk posed by unpredictable weather conditions. These measures include the following: (1) When heavy rains come suddenly: This rain often breaks the stems of rice that are ready to be harvested, specifically those with tall stems. The fallen rice stalks must be tied up in groups to stand upright. (2) When strong winds accompany the dry season: This climate change causes drought, and rice ripens prematurely (*masak sadari*). Strong winds knock the seeds to the ground and the only solution is to harvest rice ahead of time. The dry season with strong winds is the main enemy of farmers, specifically in recent years when strong winds have become unpredictable. (3) Extreme rains fall continuously every day for a long duration, sometimes from morning to noon or noon to evening: Harvesting during the rainy season slows down post-harvest work and can cause damage to rice. Leaving the paddy in the field for a long time causes shedding and crop failure. Meanwhile, harvested rice that is collected in one place may begin to sprout. This occurs because prolonged exposure to water over several days can cause the rice to regrow. This situation forces farmers to return the paddy home in a dirty state. Cleaning, separating the grains and drying the paddy are carried out in the courtyard of the house in a limited and slow manner.

4. Conclusions

This study highlights the ways in which rice farmers in the Lake Toba highlands of Indonesia—both irrigated and non-irrigated—have responded to the impacts of climate change. Farmers' awareness of climate change is primarily shaped by local experiences and traditional knowledge, which focus on extreme weather events such as excessive rainfall, declining river discharge, prolonged dry seasons, and shifting wind patterns. However, as climate variability becomes more complex and less predictable, traditional knowledge alone is no longer sufficient. Farmers increasingly rely on personal experience and trial and error to develop adaptation strategies that are often reactive rather than preventive. This study also shows that adaptation behavior differs significantly between irrigated and non-irrigated farmers due to differences in water access, topography, and infrastructure. Irrigated rice farmers tend to respond to climate stressors by improving crop maintenance,

adjusting planting times, and enhancing water management practices. In contrast, non-irrigated rice farmers—especially those whose land is located in upland areas far from Lake Toba—face greater limitations. Many of them are unable to use water pumps due to the steep terrain and are entirely dependent on rainfall. As a result, these farmers are more likely to shift to short-harvest crops like corn, which are more resilient to drought and have shorter growing cycles.

Furthermore, the unequal distribution of water resources has triggered social jealousy between farmers, particularly during the prolonged dry season. Water conflicts occur when farmers divert limited water sources to their own fields, sometimes guarding the channels overnight to secure access. These tensions reflect deeper vulnerabilities in community resilience and agricultural equity. Given these findings, there is a critical need for more inclusive government intervention. Support programs should move beyond the provision of agricultural inputs such as fertilizers and seeds to include the dissemination of accurate climate information, training in climate-resilient agricultural techniques, and infrastructure development that accounts for geographic disparities. Integrating scientific climate models with local traditional knowledge will be essential to build adaptive capacity and ensure long-term food security. Only through a hybrid approach that respects both modern science and indigenous wisdom can farming communities in the Lake Toba highlands build resilience to the ongoing and future impacts of climate change.

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Institutional Review Board Statement: All social research conducted at our university adheres to internationally recognized ethical standards, particularly those established by the International Sociological Association (ISA). The author is an active member of ISA and fully complies with its Code of Ethics. Although ethical approval for this study was granted retrospectively, we would like to emphasize that the research followed ethical procedures in line with both our institutional policies and the ISA guidelines throughout the entire process. To fulfill the journal's publication requirements, we are attaching an official letter of ethics compliance from our institution. This letter confirms that the research was conducted according to established ethical standards and that our institution recognizes and permits such research under this framework.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

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